

A microfluidic based in vitro model of synaptic competition



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ABSTRACT

Synaptic competition is widely believed to be central to the formation and function of neuronal networks, yet the underlying mechanisms are poorly described. To investigate synaptic competition in vitro, we have developed a novel two input pathway competition model using a 3-compartment microfluidic device. Axons from cultured rat cortical neurons from two different lateral compartments (inputs) innervate a common neuronal population in a separate central compartment. Inhibiting one input's activity, using the GABA_AR agonist muscimol, resulted in increased synapse numbers and axon elongation of the opposing untreated (uninhibited) inputs in the central compartment. Time lapse imaging revealed that uninhibited inputs outgrew and outconnected their inhibited counterparts. This form of competition occurs during a sensitive period ending prior to 21 DIV and is NMDAR and CamKII dependent. Surprisingly, this form of plasticity was dependent on the age of the center compartment neurons but not of the competing inputs.

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Introduction

In the developing brain, neurons undergo a substantial number of morphological and synaptic changes (Hua and Smith, 2004; Katz and Shatz, 1996). A constant balance of synapse formation, stabilization and elimination is required for the proper development and formation of neural circuits (Hua and Smith, 2004). A major contribution to these mechanisms is driven by the experience/activity/input that the brain receives during development (Hensch, 2005). One extensively studied example of this phenomenon occurs in the development of the visual cortex. When one eye is closed during development, electrophysiological observations show increases in activity and territory in favor of the open eye (Hensch, 2004; Hubel and Wiesel, 1970). These changes are also readily observable with anatomical techniques which show wider ocular dominance columns associated with the non-deprived eye than those related to the deprived eye (Gordon and Stryker, 1996; Wiesel and Hubel, 1963). The mechanism by which neural activity shapes neural circuitry during development remains a key question to be resolved in neuroscience research.

While visual cortex plasticity is one of the more well studied models of neural development, it is not without its own difficulties. Because of

the complexity of the mammalian brain, it is difficult to observe changes on cellular and molecular scales. The use of primary neuronal cultures allows for easier observation and genetic manipulation of neurons (Gartner et al., 2006; Okabe et al., 2001; Zeitelhofer et al., 2009). However, changing the activity of neurons in culture is limited to nonselective chemical treatment of the entire culture or genetic modification of a small population of neurons within the culture (Adesnik et al., 2008; Burrone et al., 2002; Harms et al., 2005; Lee et al., 1998; Lu et al., 2001; Otmakhov et al., 2004).

Microfluidic technology now provides us with the ability to grow neurons in distinct chemical environments. Microfluidics is a field of study utilizing minute volumes to control and manipulate fluid movement to create chemically isolated environments (Park et al., 2006; Whitesides, 2006). By manipulating the volumes of the individual compartments, distinct chemical environments can be created due to the hydrostatic forces present between compartments. Park et al. (2006) fabricated a two compartment microfluidic device in which two neuronal groups could remain chemically isolated but were still allowed to interact with each other through their axonal projections. Here, we use a three compartment microfluidic device to create a new dual input model to study in vitro synaptic competition.

Results

Using a 3-compartment microfluidic device, which is a specialized version of the multicompartiment chamber first developed by Park

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et al. (2006), neuronal populations were grown separately in three chemically isolated environments. Microgrooves between the compartments allowed only axons to traverse between the compartment barriers, while fluidic pressure prevented the exchange of solutions between compartments (Fig. 1) (Park et al., 2006). To distinguish the source of afferents from the two lateral input compartments, neurons in each compartment were differentially labeled by transfection with YFP- or mcherry-tagged synaptophysin (Fig. 1b) prior to plating (AMAXA nucleofection). At a later age, the GABA_AR agonist, muscimol, was used to reduce the activity of neurons in only one lateral compartment (Fig. 1a) while the other two compartments remained untreated.

Muscimol induced inhibition causes a shift in synapse distribution

To ensure that chronic inhibition with muscimol did not lead to desensitization of the GABA receptors, acute (Fig. 2a) and chronic (Fig. 2b) inhibition of neuronal activity with this treatment was confirmed by electrophysiological whole-cell patch clamp recording of cortical cells in standard culture conditions. In all cases, spontaneous network burst firing was rapidly eliminated by muscimol treatment, whereas quantal glutamate transmission and synaptic connectivity were retained. These changes remained constant even when assessed 48 h later. The data demonstrate that muscimol treatment silences all synchronous activity while synaptic connectivity remains intact.

At DIV 12–14, muscimol (10 μ M) was applied to one lateral compartment for 48 h. Following muscimol treatment neurons were fixed (1% PFA followed by methanol) and the center compartment (target) neurons were immunostained for the postsynaptic marker, PSD-95 (Affinity BioReagents). Axons from the muscimol treated compartment formed

fewer synapses (as defined by colocalized synaptophysin and PSD-95 clusters) than axons from the untreated compartment (Fig. 3, 15.41 ± 2.40 and 28.69 ± 4.19 synapses/field respectively; *t*-test, $p < 0.0078$, $n = 32$). Control cultures, in which neither lateral compartment was treated with muscimol, had synapse numbers similar to those from the muscimol treated compartment (Fig. 3a, ANOVA: $p < 0.0001$, Bonferroni post-test, where $n_{\text{control}} = 90$). This surprising result indicates that treatment of one input with muscimol results in enhanced synapse numbers from the opposing compartment's inputs.

To identify potential mechanisms underlying this difference, we interfered with several signaling pathways only within the center compartment (Fig. 4a). NMDARs are central players in activity dependent developmental competition due to their role as “Hebbian” coincidence detectors (Cull-Candy, 2004) and their subunit composition changes as neurons mature (Liu et al., 2004). Application of the NMDAR antagonist, APV (50 μ M), specifically to neurons in the center compartment (Fig. 4c, left), prevented the muscimol induced synaptic shift (muscimol treated: 12.13 ± 2.20 synapses/field, untreated: 12.29 ± 1.57 synapses/field; *t*-test: $p = 0.95$, $n = 38$). NMDAR activity leads to the activation of CamKII (Bayer et al., 2001) which in turn results in AMPAR phosphorylation and enhanced insertion into the postsynaptic membrane (Malinow and Malenka, 2002); a candidate mechanism for increases in synaptic weighting such as LTP. Treatment of the center compartment neurons with a CamKII inhibitor, KN-93 (1 μ M) (Fig. 4c, middle), also prevented the competitive effects induced by muscimol application (muscimol treated: 11.14 ± 1.79 synapses/field, untreated: 12.32 ± 1.33 synapses/field; *t*-test: $p = 0.60$, $n = 37$). Previous studies have reported a link between long term depression and synapse elimination (Bastrikova et al., 2008) and prevention of GluA2 receptor internalization

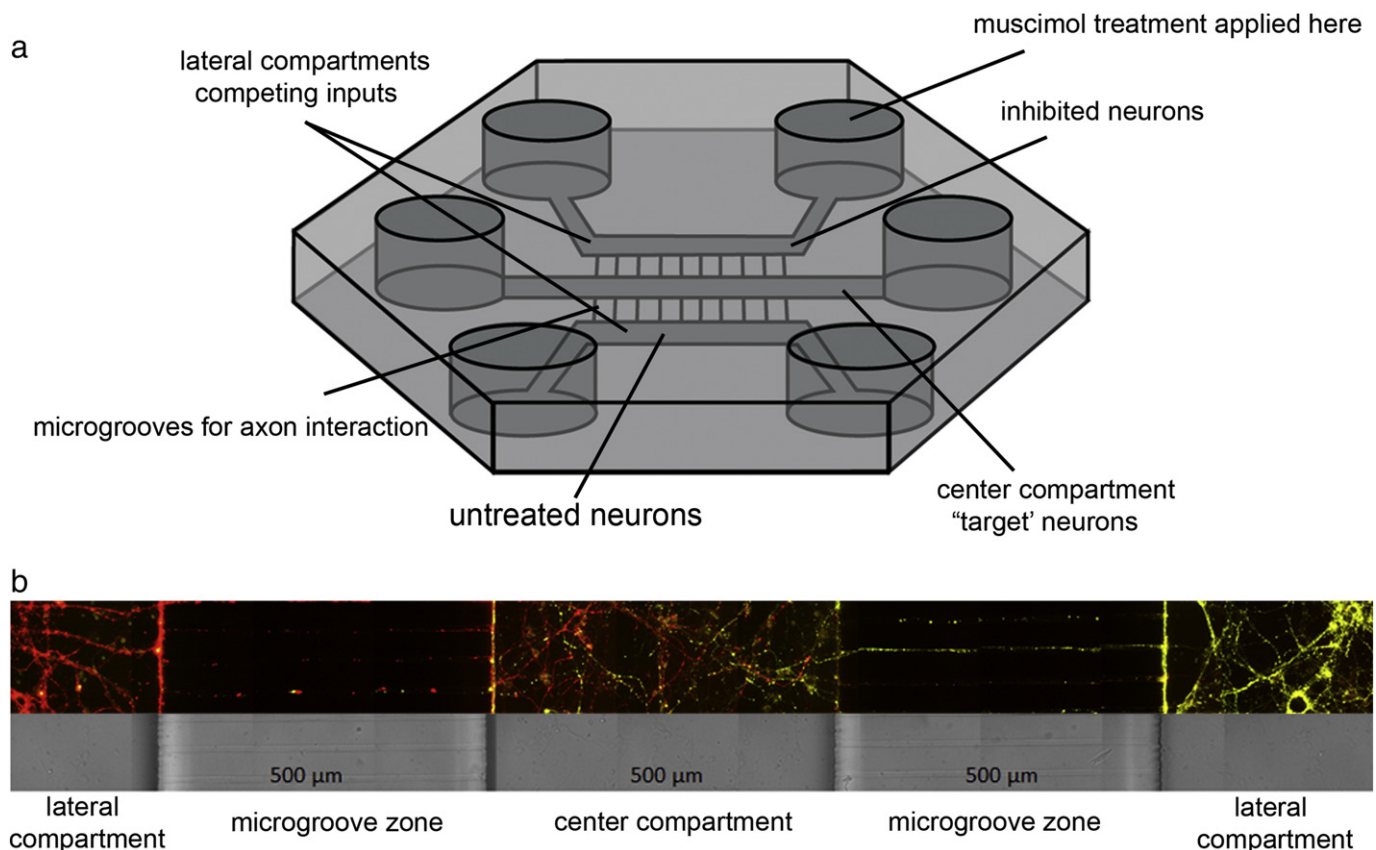


Fig. 1. Experimental set-up: a) illustration of the 3 compartment microfluidic device competition experiment. Each pair of wells is connected by a narrow channel to form a single compartment. Embryonic neurons are placed into the wells and allowed to flow through into the channels. b) Magnified view of the 3 compartment chamber viewed directly from above (center of the device, perpendicularly across the three channels). The compartment channels are separated by a series of microgrooves whose width and length allow only for axonal projections to pass through into the adjacent compartments. The image (40 $\times$ magnification) shows that only axons pass through the intercompartment microgrooves and into the next compartment while the cell bodies remain in their original chamber. The center compartment houses the “target” neurons (unlabeled). The lateral compartment’s neurons are differentially labeled with two different fluorescent proteins for purposes of identification of the compartment of origin.

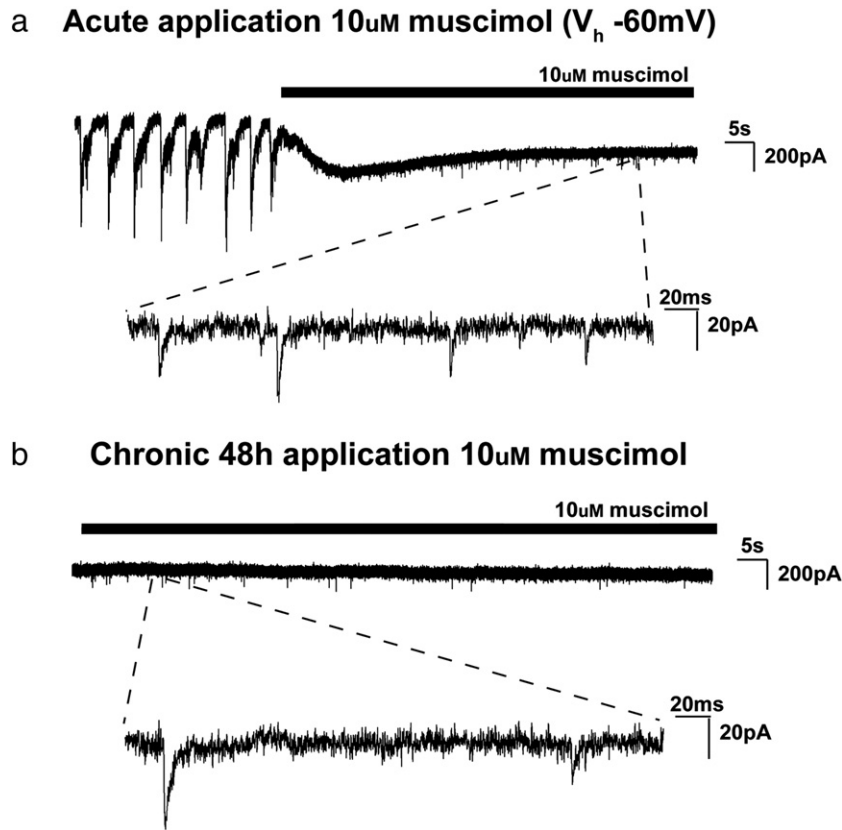


Fig. 2. 10 μ M muscimol treatment results in sustained inhibition. a) Example traces of whole-cell patch-clamp recording of cortical neurons showing inhibitory silencing of spontaneous synaptic burst activity immediately after muscimol (10 μ M) application (DIV 12) and b) sustained inhibition 48 h later (DIV 14).

has been shown to block LTD (Ahmadian et al., 2004). Furthermore, GluA2 expression levels correlate with synapse number, size, density and stability in vitro (Passafaro et al., 2003; Ripley et al., 2010; Saglietti et al., 2007). However, application of the TAT-GluR2_{3y} peptide, which has been shown to prevent GluA2 internalization (Ahmadian et al., 2004), to the center compartment neurons (Fig. 4c, right) did not prevent the competitive synaptic changes that we observed (muscimol treated: 16.90 ± 2.04 synapses/field, untreated: 26.62 ± 2.95 synapses/field; *t*-test: *p* = 0.0083, *n* = 42). Fig. 4b shows that as in our previous results (Fig. 3a), when a change in synapse number occurred it resulted from an increase in the number of synapses from the untreated input compartment.

Surprisingly, no significant difference in synapse density (defined as the number of synapses per 10 μ m of axon length) between the treated (0.210 ± 0.041 synapses/10 μ m, *n* = 25) and untreated (0.309 ± 0.058 synapses/10 μ m, *n* = 25) inputs was observed (ANOVA = 0.33; control: 0.265 ± 0.36 synapses/10 μ m, *n* = 33; Fig. 5a). This suggests that the observed difference in synapse number between treated and untreated inputs might be due to differences in axonal length and, therefore, growth. To examine this point directly, we assessed axonal extension and motility using time lapse imaging at 30 min intervals. The overall motility of axons (observed over 10–20 h during the muscimol treatment period) originating from the treated and untreated lateral compartments in the center compartment was classified into three categories: advancing, retracting and unchanged. At this age (DIV 12–14), independent of treatment, both sets of lateral compartment axons showed a net positive growth. However, the degree of axonal growth was different for the two incoming groups of axons. Afferents originating from the untreated compartment (9.25 ± 0.88 μ m/30 min, *n* = 11) on average moved twice as fast and farther than those from the muscimol treated compartment (4.99 ± 0.58 μ m/30 min, *n* = 11). The average growth rate of axons from the muscimol

treated compartment was not significantly different from that of axons in cultures in which neither lateral compartment was treated with muscimol (6.34 ± 0.73 μ m/30 min, *n* = 17). Rather axons from the untreated compartment increased their outgrowth and synapse number when opposed by muscimol treated axons (Fig. 5b, ANOVA *p* = 0.0025, Bonferroni post-test). The surprising result is that *untreated* inputs undergo enhanced axon growth in a competitive environment when the activity of a competing group of inputs is reduced by muscimol application.

Changes in synapse distribution induced by muscimol no longer occur by 3 weeks in vitro

To determine whether developmental age plays a role in this process, 48 h muscimol treatments were initiated at DIV 8, 19 and 26 (Fig. 6b–c, d–e, f–g, respectively). Muscimol treatment from DIV 8–10, led to a significant difference in the synapse number between treated and untreated lateral compartment axons (muscimol treated: 3.14 ± 0.71 synapses/field; untreated: 6.60 ± 0.67 synapses/field; *t*-test: *p* = 0.0007; *n* = 35). However, there were no effects at the later ages of DIV 19–21 (muscimol treated: 16.56 ± 2.62 synapses/field; untreated: 20.86 ± 2.79 synapses/field; *t*-test *p* = 0.26, *n* = 36) or DIV 26–28 (muscimol treated: 20.38 ± 2.75 synapses/field; untreated: 17.71 ± 1.78 synapses/field; *t*-test: *p* = 0.42, *n* = 34). These experiments demonstrate that the age of the cultures is an important variable in this model (Fig. 6a) and suggests the existence of a sensitive period.

The age of the neuronal ‘target’ group is a critical factor for the muscimol induced changes

To isolate the role of incoming afferents versus target cells in this age dependent process, two experiments were performed in which the

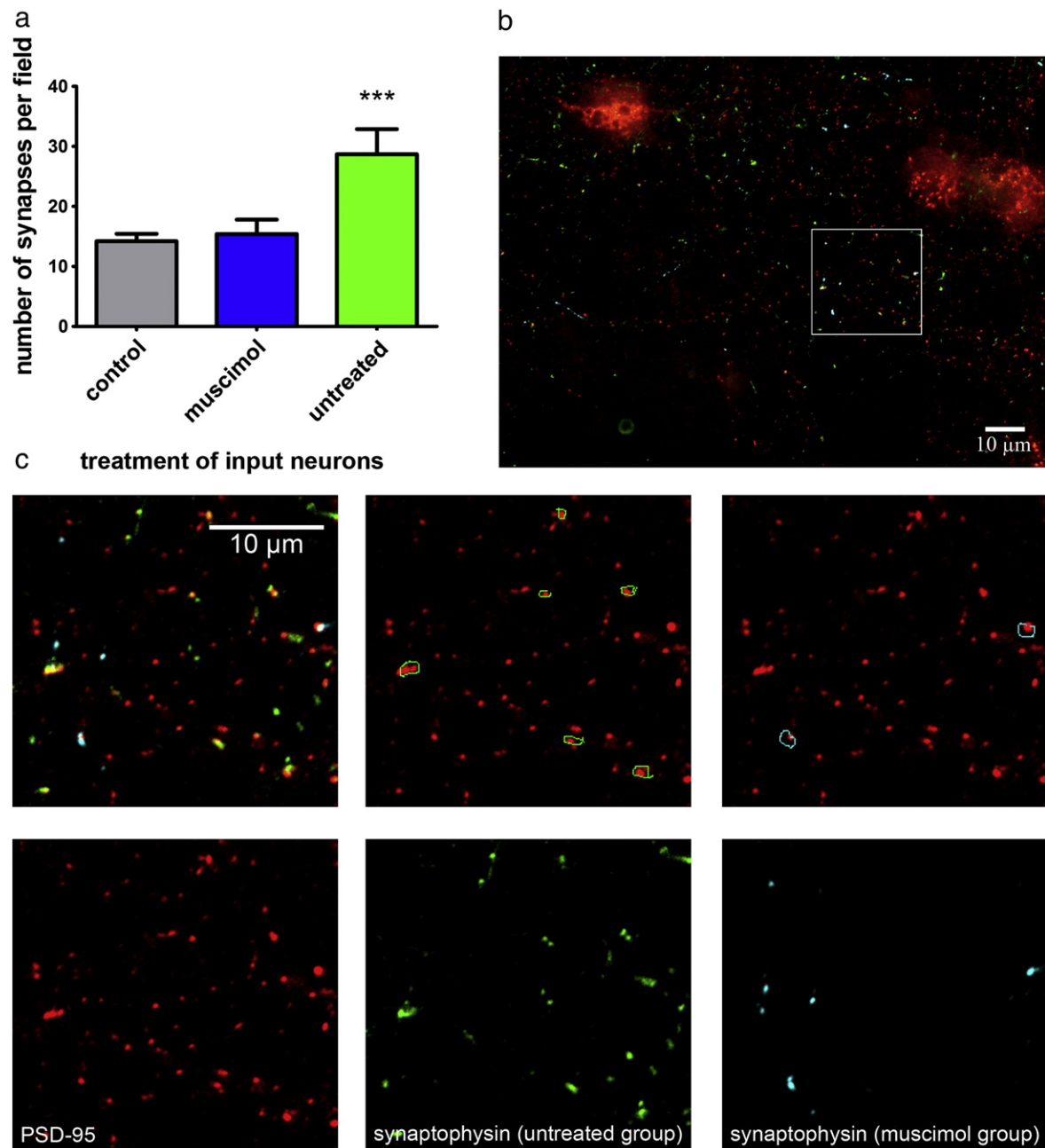


Fig. 3. Differences in activity between inputs result in a change of synapse numbers in the center compartment: a) shows that muscimol treatment of one lateral compartment leads to increases in synapse number for the opposing (untreated) lateral compartment's inputs, ANOVA $p < 0.0001$, Bonferroni post-test. b) Example of a DIV 14 image taken from the center compartment in which one lateral compartment was treated with muscimol. All images were taken at the middle of the center compartment to prevent proximity bias for lateral inputs. Red indicates PSD-95, green indicates afferents from the untreated lateral compartment, and blue indicates afferents from the muscimol treated lateral compartment for all images. c) Cropped section of b. Top, left to right: merged image of treated and untreated afferents and PSD-95; PSD-95 with outlines (green) of overlapping synaptophysin originating from the untreated lateral compartment; right: PSD-95 with outlines (blue) of overlapping synaptophysin originating from the muscimol treated lateral compartment; Bottom, left to right: separated images of PSD-95, synaptophysin from the untreated lateral compartment and synaptophysin from the muscimol treated lateral compartment respectively.

lateral compartment inputs and the center compartment neurons differed in developmental age by 2 weeks (inputs DIV 12 or 26 and vice versa for center compartments, Fig. 7a). Using muscimol to reduce neural activity of one lateral compartment, as before, no difference in synapse number was observed between treated and untreated inputs when the “target” neurons were two weeks older than the input axons. However, when the lateral compartment axons were treated from DIV 26–28 and the “target” neurons were 2 weeks younger (DIV 12–14), fewer center compartment synapses were formed by afferents from the muscimol treated compartment than by the untreated inputs (Fig. 7b, ANOVA $p < 0.0001$, Bonferroni post-test). These results

demonstrate that muscimol induced changes depend on the age of the center compartment (target) neurons. In addition, these changes showed a similar pattern to that seen in Figs. 3a and 4b using neurons of the same age. Again, the difference in synapse number appears to result from an increase in synapses of the untreated competing inputs. Furthermore, these changes can still occur when the input axons were beyond the ages at which these effects normally occur, provided that the center compartment (target) neurons were at the appropriate age. Interestingly, we also observed in these heterochronic situations that “older” postsynaptic targets would readily form more synapses with younger presynaptic afferents compared to the reverse situation.

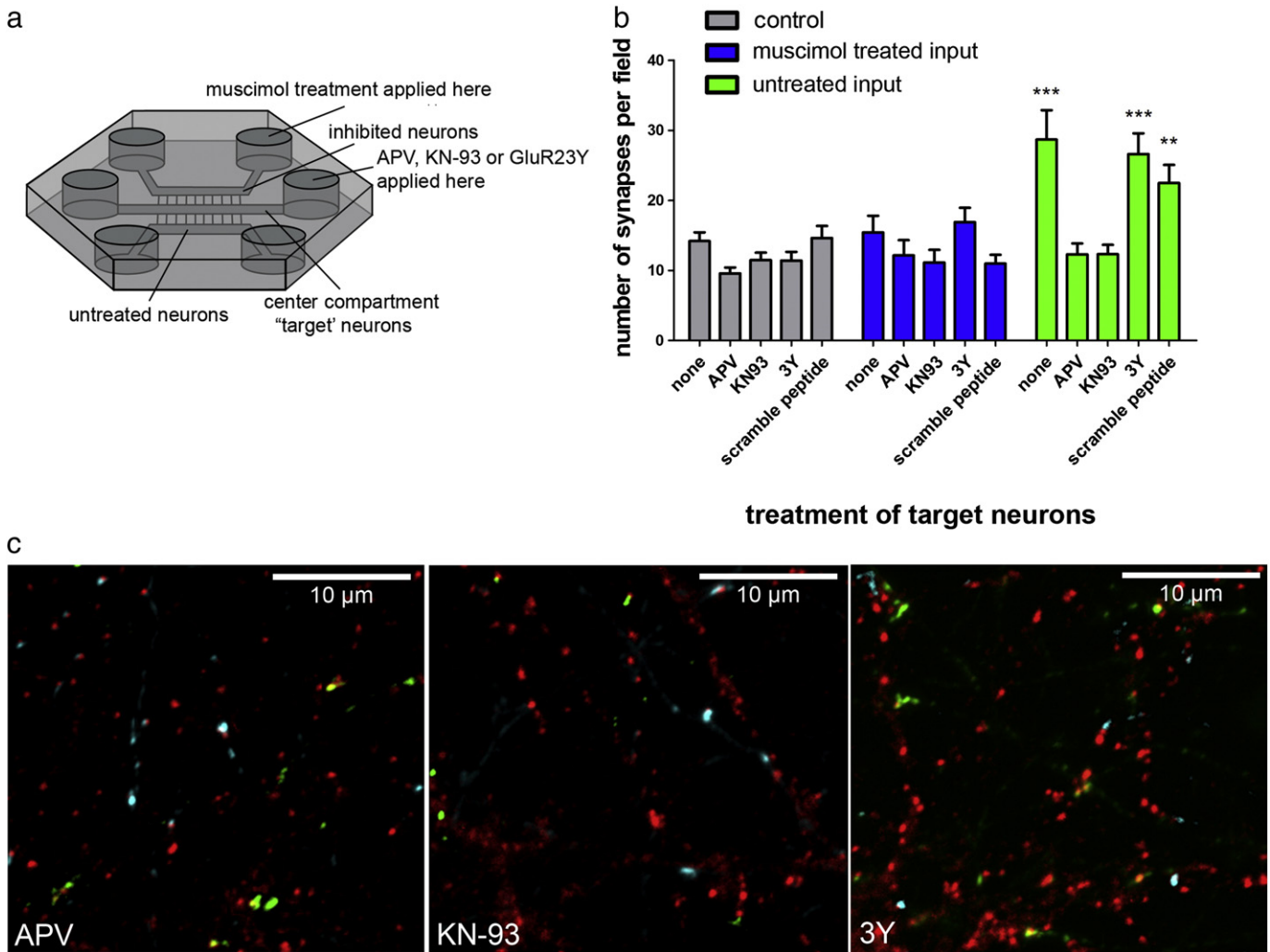


Fig. 4. Pharmacological treatments applied to the center ('target' neurons) compartment can prevent muscimol induced competition. **a)** Illustration of the 3 compartment competition experiment with various pharmacological treatments in the center compartment. **b)** Comparison of the number of synapses per field of all treatments, ANOVA $p < 0.0001$, Bonferroni post-test ($n = 90, 32, 32, 70, 38, 38, 72, 37, 37, 42, 42, 63, 63, 40, 36$ in order of appearance on the graph). This panel shows that treating the center compartment with the 3Y peptide, a scrambled peptide control or leaving it untreated results in increased synapse numbers from the untreated lateral compartment when the other compartment is treated with muscimol. Note that the muscimol treatment does not reduce the number of synapses/field relative to control conditions in which both lateral compartments remain untreated. Rather the effect is to increase synapse numbers from the untreated compartment. This effect is blocked by treatment with APV or KN-93 in the center compartment. **c)** Cropped images of observed fields and colocalization of inputs and PSD-95 staining within the center compartment of cultures treated (in the center compartment) with either APV, KN-93 or 3Y peptide in which one lateral compartment was treated with muscimol (blue) and the other remained untreated (green).

However, whether this occurs due to an actual age preference or whether it is simply the case that the older presynaptic neurons would just have more readily available synaptic targets remains to be determined.

Discussion

We used a 3-compartment microfluidic device to create a novel in vitro dual input synaptic competition model. Reducing the activity of inputs from one lateral compartment, while leaving the other unchanged, results in increased synapse numbers associated with the untreated input afferents. These effects are due primarily to untreated axons advancing further/more rapidly in the center (target) compartment than muscimol treated axons as opposed to increases in synapse density along the axon. One possible explanation is that the stabilization of presynaptic markers such as synaptophysin correlates with further axonal branching and growth (Ruthazer et al., 2006) and these sites are more likely candidates for becoming stable presynaptic terminals for synapses (Sabo et al., 2006). Therefore, the increased axonal growth

in our model may be due to the increased stability of synaptic connections on 'winning' afferents which then promotes further axonal branching and growth.

The mechanisms underlying these changes involve NMDAR activation and CamKII dependent signaling; however, further work is required to determine whether LTP-like AMPAR membrane insertions play a significant role in this competition model. Furthermore, the changes appear independent of GluR2 internalization and LTD-like processes. LTD is associated with synapse elimination and retraction (Bastrikova et al., 2008) while our model appears to predominantly involve increased growth. L-type calcium channels have also been shown to have a role in activity dependent synaptic plasticity (Goold and Nicoll, 2010; Wang et al., 2011; Wu et al., 2008). Future studies examining their role will help to further our understanding of synaptic competition.

Previous experiments have shown that synapse formation and maturation in cultured neurons occur most rapidly at approximately two weeks in culture (Craig et al., 1993). The effects observed here were found to be age dependent, suggesting a sensitive period akin to

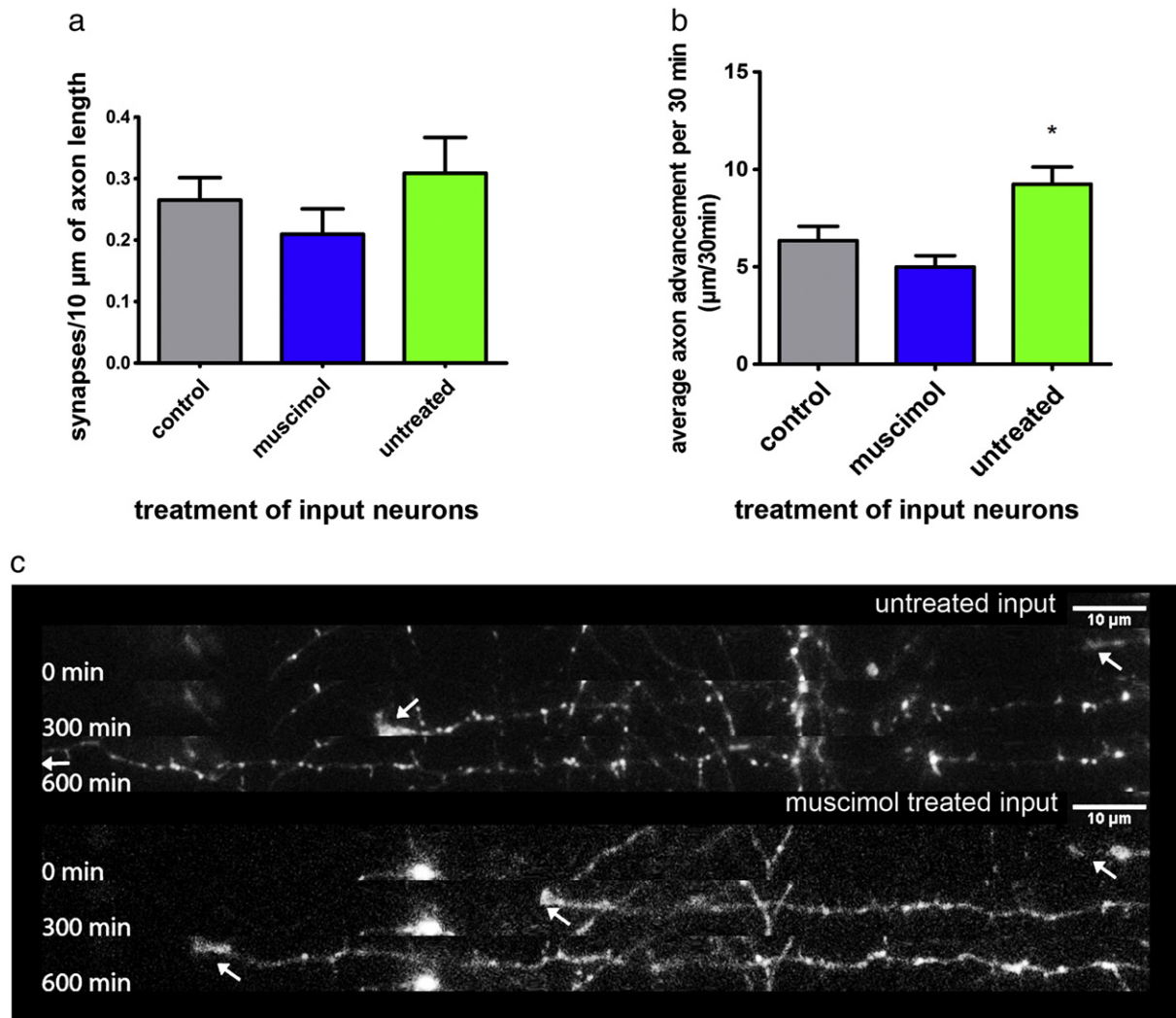


Fig. 5. a) Comparison of synaptic densities per 10 μm of axon, ANOVA: $p = 0.33$. Despite the marked changes in synapse numbers among the muscimol treated and untreated inputs, the density of synapses per unit length remained the same for the treated and untreated compartments' inputs. b) Comparison of axon growth rate, ANOVA: $p = 0.0025$, Bonferroni post-test. The untreated axons show greater/faster growth. c) Time lapse image of axon growth of treated and untreated afferents in the 'target' compartment over 10 h.

the *in vivo* critical period. This sensitive period was specifically dependent on the age of the center compartment (target) neurons and not the age of the lateral compartment (input) neurons. Because this model is dependent on the age of the center compartment neurons, further investigations into the postsynaptic role of synapse selection are required. In addition, center compartment axonal projections will also form synapses with lateral compartment dendrites, and the roles that these connections play may also provide valuable insight into this model of competition.

Our study has demonstrated that decreased neuronal activity within one population of incoming afferents can influence the synapse formation and growth of axons of a competing neuronal population. The experimental model we introduce here has many potential uses in the study of activity dependent synaptic plasticity.

Experimental methods

Microfluidic devices

Microfluidic devices were constructed as detailed in (Park et al., 2006) with an additional center compartment 0.5 mm in width. Devices were soaked in PBS, rinsed in water and dried prior to assembly. Devices were then placed onto No. 1.5 glass coverslips. Assembled devices were filled with poly-D-lysine and kept at 37 °C for at least 24 h. Devices were

then washed with distilled water and kept at 37 °C (while compartments were still filled with water) until use.

Cell culture and treatment

All animal experiments were approved by the Animal Care Committee of the University of British Columbia. Primary cortical neuronal cultures were prepared from Wistar rats' (University of British Columbia Animal Care Centre, Vancouver, BC, Canada) brains at embryonic days 18–19. Cultures were maintained in Neurobasal medium supplemented with B27, 0.5 mM Glutamax (Invitrogen) and Penicillin–Streptomycin (Invitrogen).

Transfections were performed via nucleofection at DIV 0 using plasmids containing the pLL3.7 backbone with synapsin promoter and expressing synaptophysin-YFP or -mcherry. Freshly dissociated neurons (4 million) were pelleted for nucleofection using an Amaxa Nucleofector II (Lonza) program O-003 with 2 μg of plasmid DNA and then immediately resuspended in 1 mL DMEM (Invitrogen) with 10% FBS (Invitrogen) for plating.

Plating of neurons into the microfluidic devices was done by placing 10 μL of the neuronal cell suspension (4 million cells/mL for transfected neurons, 3 million cells/mL for nontransfected) into each compartment of the device. Cells were allowed to flow through the compartment and

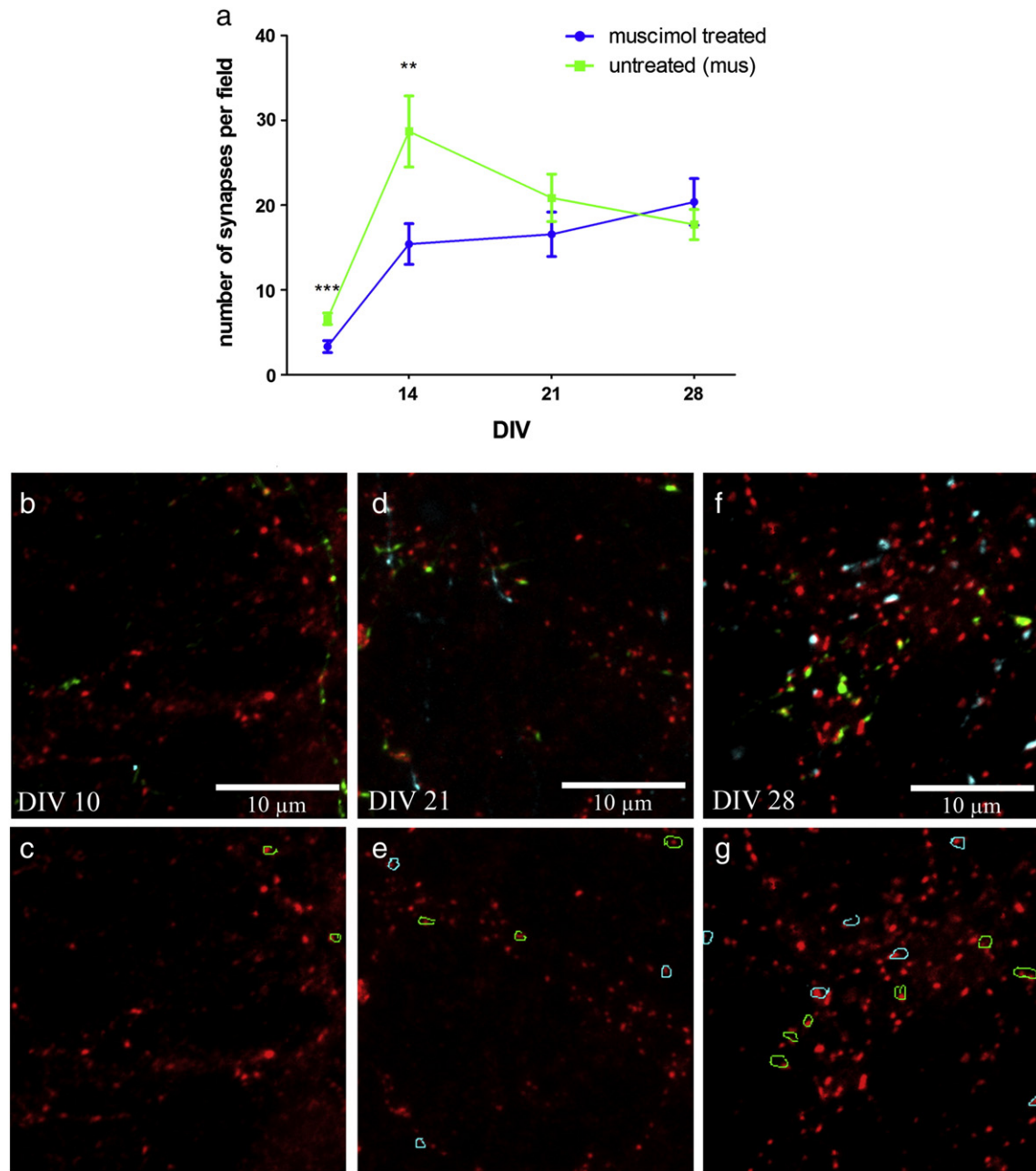


Fig. 6. The developmental age (in vitro) of the neurons is an important contributor to the effects observed: a) shows the effects of varying the age at which muscimol was applied to one lateral compartment. Changes in synapse density between treated and untreated lateral inputs were observed at DIV 8–10 and 12–14 but not at later ages. b–g) Cropped images of the center compartment in which one lateral compartment was treated with muscimol for 48 h then imaged at DIV 10 (b, c), 21 (d, e), and 28. (f, g) Cultures respectively (the puncta representing untreated inputs are in green, muscimol treated inputs in blue. Images on the top are merged images. Images on the bottom highlight where the colocalization occurs).

wells were topped up with Neurobasal medium once the neurons settled.

After plating and before fixation, compartments were qualitatively assessed to ensure neuronal densities in each channel compartment were similar. We counterbalanced muscimol treatments between mcherry and YFP labeled compartments to rule out any effect of fluorophore composition in our results.

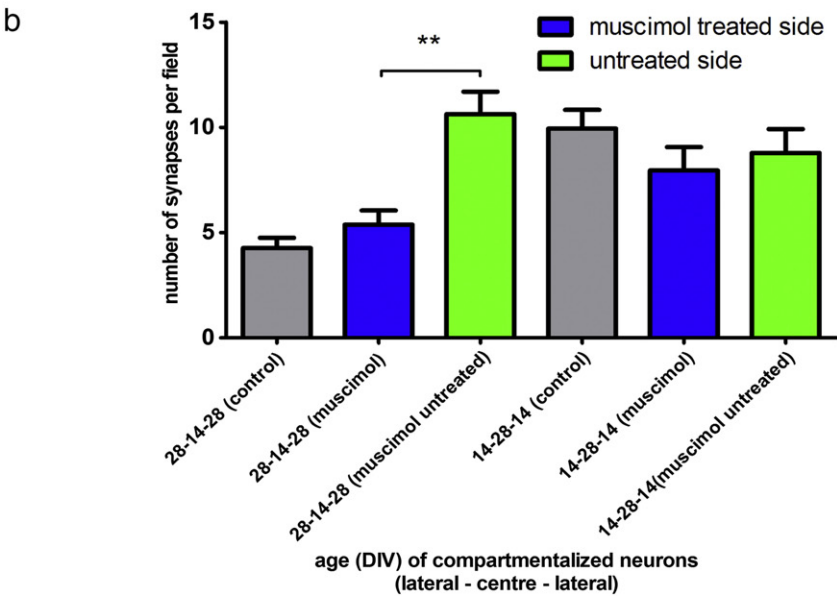
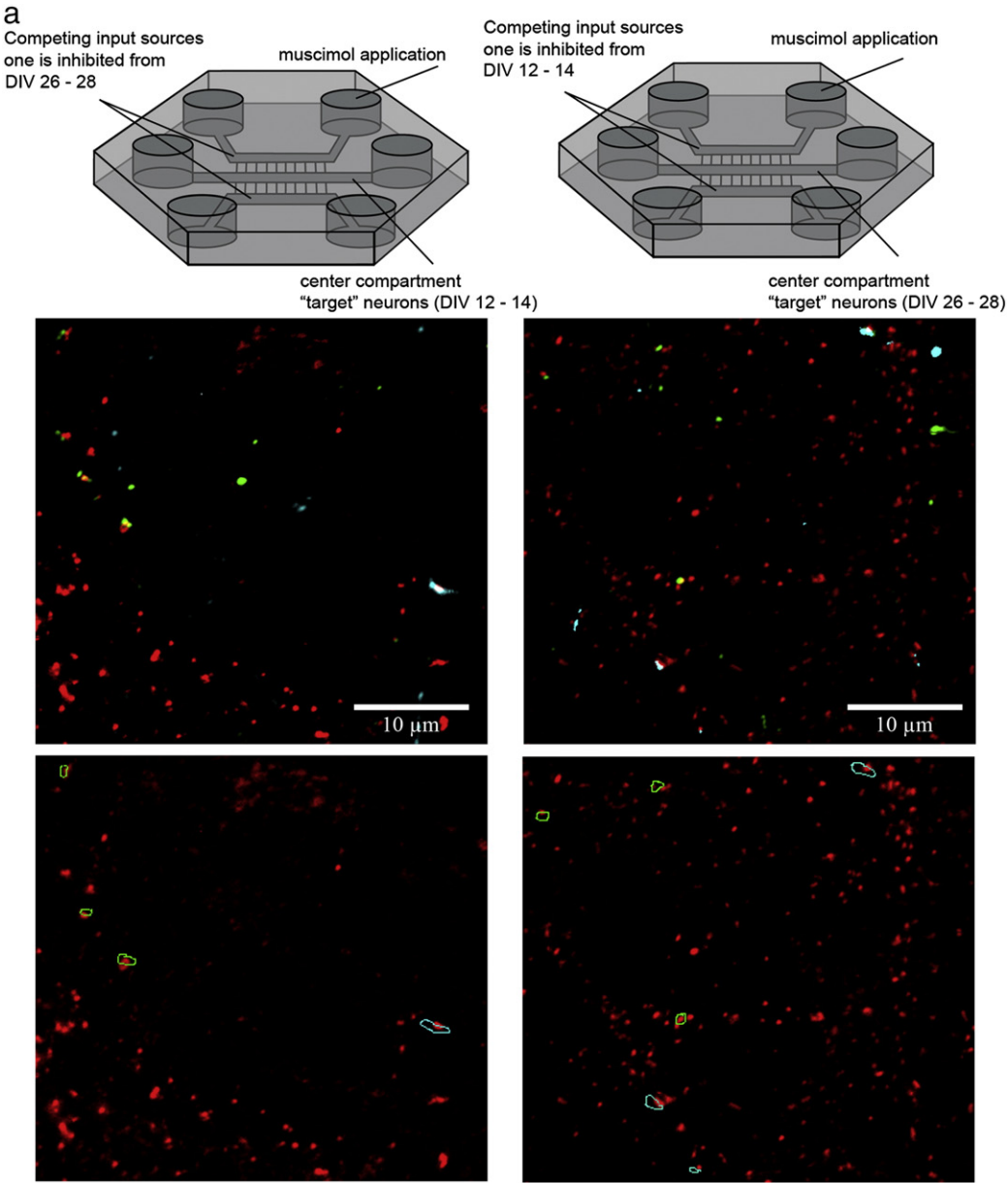
Treatments

Tests using fluorescent dyes (fluorescein or cresyl violet) to fill one lateral compartment (120 μ L total volume of dye) showed that after 48 h a dye concentration of less than 1% was found in the other 2 compartments (with total volumes of 600 μ L each). When the center compartment was filled with a total dye volume of 250 μ L, there was a

$4.1 \pm 4.3\%$ concentration of dye found in a lateral compartment containing a total volume of 120 μ L of water. The other lateral compartment containing a total volume of 600 μ L of water had a dye concentration of less than 1% after 48 h. %concentration was based on starting dye concentration of the filled well (100 μ M fluorescein or crystal violet). We maintained conditions in the experiments reported here that would allow for less than 1% muscimol leakage into adjacent compartments during the 48 h treatment period.

Immunocytochemistry

Neurons were quickly washed with PBS, then fixed with 1% PFA for 4 min followed immediately by methanol (-20°C) for 8 min, then washed with PBS again. Neurons were then stained for PSD-95 (Affinity



BioReagents, 1:1000 dilution in 3% BSA in PBS) overnight followed by secondary antibody staining (anti mouse Alexa 647, Invitrogen, 1:1000 dilution in 3% BSA in PBS) for 1 h. Coverslips were then mounted on slides using Fluoromount-G and left at room temperature overnight before imaging.

Imaging microscopy and analysis

Image acquisition

Images were acquired on a Zeiss Axioplan2 microscope with a $63\times$ 1.4 numerical aperture oil objective and Photometrics Sensys cooled CCD camera using MetaVue imaging software (Molecular Devices) and customized filter sets (immunocytochemistry).

Images were acquired on a Zeiss AxioObserver microscope with a $63\times$ 1.4 numerical aperture oil objective (immunocytochemistry) or $40\times$ 0.95 numerical aperture air objective (live imaging) and Zeiss HRM camera using Axiovision imaging software (Zeiss) and customized filter sets.

For immunocytochemistry experiments n equals the number of field images taken from at least 3 independent experiments. All fields were taken at the center line of the 'target' compartment to prevent any proximity bias between lateral afferent compartments. A field is defined as the field of view observed under the microscope (i.e. 1 image taken under $63\times$ magnification for the immunocytochemistry experiments).

For time lapse image experiments, n equals the numbers of axons observed from at least 2 independent experiments. All time lapse imaging experiments used observations at the center line of the center compartment to prevent any proximity bias between lateral afferents.

Colocalization image analysis was performed using the ImageJ colocalization plugin found on the ImageJ website (<http://rsbweb.nih.gov/ij/index.html>). Before running the plugin, images were (blindly) thresholded using ImageJ and binary masks were made to include only puncta 5–300 pixels in size (1 pixel = 0.1 μ m). Axon segments were (blindly) selected under the criterion that segments were clearly distinguishable as a single axon from a single source.

Growth classification analysis was done by identifying all axon growth cones in the field of view. Growth cones were observed over the duration of the imaging or until outside the field of view.

Axon growth rate was obtained by identifying and selecting axon growth cones within the field of view 4–24 h into treatment. Axons were observed until imaging concluded or axons moved out of the field of view. Axon growth segments were (blindly) selected under the criterion that segments were clearly distinguishable as a single axon from a single source.

All imaging values are described by mean $\pm$ SEM.

Electrophysiology

Whole-cell patch-clamp recordings were performed on CTX cells at DIV 12–14, voltage clamp recordings were obtained at $V_h = -60$ mV and the membrane test function was used to determine intrinsic membrane properties ~ 1 min after obtaining whole-cell configuration, as described previously (Kaufman et al., 2012; Milnerwood et al., 2012; Tapia et al., 2012). Briefly, neurones were perfused at room temperature with extracellular solution (ECS) containing (in mM unless stated): 167 NaCl, 2.4 KCl, 1 MgCl₂, 10 glucose, 10 HEPES, 2 CaCl₂,

pH 7.3, and 300 mOsm. Pipette resistance (R_p) was 3–6 M Ω when filled with (in mM): 130 Cs methanesulfonate, 5 CsCl, 4 NaCl, 1 MgCl₂, 5 EGTA, 10 HEPES, 5 QX-314, 0.5 GTP, 10 Na₂-phosphocreatine, and 5 MgATP, 0.1 spermine, pH 7.2, 290 mOsm. Data were acquired by Multiclamp 700B amplifier and signals were filtered at 2 kHz, digitized at 10 kHz, and analyzed in Clampfit10 (Molecular Devices). Tolerance for series resistance (R_s) was <25 M Ω and uncompensated; Delta R_s tolerance was $<10\%$. Muscimol was added by bath infusion or already added to the ECS for experiments with 24 h treatments and mEPSCs were visualized using Clampfit10.

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Fig. 7. Muscimol induced competition in this model requires the center 'target' compartment neurons to be an appropriate age. a) Illustration of heterochronic experiments with (below) corresponding cropped images of the center compartment in which one lateral compartment was treated with muscimol (the puncta of untreated inputs are in green, muscimol treated inputs in blue). *Left side:* illustration of heterochronic experiment with lateral compartment neurons at DIV 28 and center compartment neurons at DIV 14, top image: merged image of treated and untreated afferents and PSD-95, and bottom image: PSD-95 from the center compartment with outlines (green) of overlapping synaptophysin originating from the untreated lateral compartment and outlines (blue) of overlapping synaptophysin originating from the muscimol treated lateral compartment. *Right side:* illustration of heterochronic experiment with lateral compartment neurons at DIV 14 and center compartment neurons at DIV 28, top image: merged image of treated and untreated afferents and PSD-95, and bottom image: PSD-95 from the center compartment with outlines (green) of overlapping synaptophysin originating from the untreated lateral compartment and outlines (blue) of overlapping synaptophysin originating from the muscimol treated lateral compartment. b) Treating one lateral compartment with muscimol creates synaptic asymmetry in the center compartment only when the postsynaptic neurons are at an appropriate age, ANOVA $p < 0.0001$, Bonferroni post-test ($n = 38, 24, 24, 34, 23, 23$ in order of appearance on the graph).

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